

Database Management System

BCA — Semester 3 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. Define serial, non-serial and serializable schedules with example. How can you test serializability of a schedule? Explain with an example

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What is Transaction? State and explain the states of transaction with transition diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. From the the relations given below, answer the following questions. EMPLOYEE (Ssn, Fname, Bdate, Address, Sex, Salary, Super_ssn, Dno) DEPARTMENT (Dname, Dnumber, Mgr_ssn, Mgr_start_date) PROJECT (Pname, Pnumber, Plocation, Dnum) DEPENDENT (Essn, Dependent_name, Sex, Bdate, Relationship) DEPT_LOCATIONS (Dnumber, Dlocation) a) Retrieve the name and address of all employees who work for the 'Computer' department in SQL b) For each Department, retrieve the department number, the number of employees in the department, and their average salary using SQL c) Retrieve the employees name and their Dname and Pname ordered by the employee's Dname d) Retrieve the name and salary of all employees who work in department number 5 using Relational Algebra e) Retrieve the name of the manager of each department using Relational Algebra.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What are the Advantages of using DBMS? Explain in brief.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is Attributes? Explain different types of Attributes with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Define relation, instance and schema with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is Normalization? Explain the 2NF with Examples.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is Shared/Exclusive (Read/Write) Locks? How it is different from binary Locks.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is Deadlock in DBMS? Explain with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is shadow paging? How it is used for database recovery?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is data independence? How three schema architecture ensures logical and physical data independence?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. What is NoSQL? Explain the characteristics of NoSQL.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.